

Portfolio Risk Dashboard

Asset Management / Wealth Management Portfolio Project | 10-Stock Model Portfolio vs. SPY

12.9% Annualized Return	20.1% Volatility	1.22 Beta	24.4% Max Drawdown
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Portfolio Overview

The dashboard analyzes a diversified U.S. equity model portfolio containing Apple, Microsoft, NVIDIA, JPMorgan Chase, Amazon, Alphabet, Exxon Mobil, Johnson & Johnson, Costco and Visa, with SPY as the benchmark. The workbook is designed to demonstrate portfolio construction, performance measurement, downside-risk analysis, concentration monitoring and diversification analytics.

Risk Analytics

Metric	Portfolio	Interpretation
Annualized Return	12.9%	Geometric annualized return
Annualized Volatility	20.1%	Total variability of monthly returns
Sharpe Ratio	0.44	Excess return per unit of volatility using 4.0% risk-free rate
Beta vs. SPY	1.22	Portfolio is more market-sensitive than the benchmark
1-Month 95% VaR	8.2%	Estimated loss threshold in the worst 5% of months
1-Month 95% CVaR	9.3%	Average loss in months beyond the VaR threshold
Maximum Drawdown	24.4%	Largest peak-to-trough decline in the scenario period

Portfolio Construction

- **Growth tilt:** AAPL, MSFT, NVDA, AMZN and GOOGL provide substantial exposure to technology and digital growth.
- **Diversifiers:** JPM and Visa add financial exposure; XOM, JNJ and Costco broaden sector exposure and reduce dependence on a single economic driver.
- **Concentration controls:** the largest individual position is 14%, while technology represents 39% of the model portfolio.
- **Diversification score:** the effective-number-of-holdings methodology produces a score of approximately 95%, indicating weights are relatively balanced.

Risk Management Takeaways

- **Correlation matters:** mega-cap growth holdings can move together during market stress, so position count alone can overstate diversification.
- **Tail risk:** VaR and CVaR quantify potential downside beyond standard volatility and help frame adverse monthly scenarios.
- **Drawdown monitoring:** maximum drawdown captures the investor experience of peak-to-trough losses that volatility can obscure.
- **Risk contribution:** the dashboard highlights which holdings contribute the most standalone volatility relative to their portfolio weights.

Recruiting / Interview Takeaway

This project demonstrates how I would evaluate a portfolio beyond simple returns. I built a dashboard that combines performance, volatility, beta, Sharpe ratio, VaR, CVaR, drawdown, concentration, correlation and holding-level risk contribution to identify where portfolio risk is actually coming from and whether the portfolio is adequately diversified.

Methodology Note

The workbook uses a 60-month deterministic illustrative return series calibrated to plausible equity volatility and beta assumptions; it is not actual historical security performance. The model is intentionally structured so verified total-return data can replace the blue input cells and automatically update all analytics. This project is for educational and recruiting purposes and is not investment advice.